# ELMERSON R

AI & Full-Stack Developer

+91 9384984937 | elmersonrajan@gmail.com | elmerson.me | linkedin.com/in/elmerson | github.com/elmersonrajan

## SUMMARY

B.Tech graduate in Artificial Intelligence and Data Science with production experience building AI-driven and full-stack products end to end — conversational voice agents, real-time WebRTC platforms and cross-platform mobile apps. Strong foundation in Python and machine learning, combined with hands-on delivery of client software from architecture through deployment.

## EXPERIENCE

### Software Developer Intern — Doxa Technology Solutions

*Aug 2026 – Present*

- **Susee Hyundai AI Calling Agent —** Built an AI voice agent to replace the dealership front desk, handling inbound customer calls end to end. Combined Gemini Flash 3.1 for conversation and intent handling with Sarvam AI and Google TTS for natural multilingual speech, deployed on AWS EC2 with Supabase as the data and call-log layer.
- **Goimomi Holidays AI Booking Bot —** Built an AI chat assistant that takes a traveller from enquiry to confirmed booking in one conversation — hotel rooms and cab reservations handled inside the chat, with Zoho Payments integrated so the customer pays without ever leaving it.
- **Thangam Stay Inn —** Developed a hotel website with an integrated online booking engine, Aadhaar-based guest verification, and Razorpay payment integration, backed by an admin CRM for managing reservations, guest records and occupancy.
- **OJK Jobs WhatsApp Bot —** Built a WhatsApp job-alert and application bot on the WhatsApp Business Platform, configured through Meta for Developers and a Facebook Business account, with CRM integration so candidate enquiries and applications flow straight to the recruitment team.

### Freelance Full-Stack Developer — Self-Employed

*2025 – Present*

**LYN MEET — Live Virtual Classroom (Web + Mobile)**

*class.lynindia.in*

Node.js · mediasoup (WebRTC SFU) · Socket.IO · React · Vite · MySQL · FFmpeg · Flutter · coturn

- Designed and built an end-to-end classroom meeting platform serving teachers and students in production, replacing third-party conferencing with a self-hosted stack.
- Implemented a mediasoup SFU for multi-party camera, microphone and screen sharing, including ICE/TURN setup (coturn), adaptive video quality, and reconnect handling that keeps a class alive through a teacher's network drop.
- Built a real-time collaborative whiteboard over Socket.IO with multiple board tabs, PDF/image import, zoom, and late-join state sync, so a student joining mid-class sees the same board as everyone else.
- Engineered a server-side recording pipeline: a single FFmpeg ingest keeps audio, camera and screen on one clock, then a compose pass renders the whiteboard/screen with a camera inset into one MP4 per session, auto-published to the learning platform for playback.
- Enforced role-based permissions server-side (teacher / coordinator / student) across mute control, screen share, drawing, polls and session close, plus SSO via one-time signed tickets validated against the platform's user directory — no client-trusted roles.
- Added automatic attendance tracking with rejoin handling, idle-room reclamation to free SFU resources, and a Node test suite whose pre-deploy FFmpeg pipeline check gates releases; shipped a companion Flutter student app on the same socket contract.

**MLCC Youth — Community App**

*play.google.com/store/apps/details?id=com.mlccapp.mlccapp*

Flutter · Firebase · OneSignal · Google Play Store

- Designed and launched a community app on the Google Play Store for a youth organisation, covering events, announcements and member content.
- Used Firebase for authentication, real-time content updates and hosting, and integrated OneSignal for targeted push notifications and event reminders.
- Managed the full release cycle — Play Console listing, store compliance, versioned rollouts and post-launch updates based on user feedback.

**David Bikes — Bike Workshop Management App**

Flutter · Firebase · PDF generation

- Built a workshop management app that digitises customer and vehicle records, letting staff register bikes against an owner and track each vehicle's service history.
- Implemented service invoicing with live invoice updates, and one-tap PDF bill generation that can be shared directly with clients from the app.

- Replaced the workshop's paper-based billing with a single searchable customer database, cutting the time to produce and send a bill.

**Mathan Jewellers — Chit Fund Register (Web + Installable App)**
*mathan-jewellers.web.app*
React · TypeScript · Vite · Firebase Firestore / Auth / Hosting · Tailwind CSS · jsPDF · PWA

- Built a bilingual (English / Tamil) chit-fund register for a jewellery business, replacing a paper ledger with searchable customer and payment records shared live across every device.
- Implemented instant client-side search across name, chit number and phone with ranked partial matching, plus a month-by-month collection feed with running totals.
- Generated branded, paginated PDF statements with repeating headers, a single closing total, and one-tap sharing to WhatsApp through the native share sheet.
- Secured the data in Firestore rules rather than the UI: access requires an allow-listed admin document, and every write is shape- and range-validated server-side.
- Separated a framework-free domain layer from the React UI so the same queries, validation and search can back a future native app; shipped as an installable, offline-capable PWA.

## PROJECTS

### AI Personalized Learning System
Python · Scikit-learn · Streamlit · Gemini LLM API

- Built an AI-powered system that analyses learner behaviour and delivers personalized educational recommendations, profiling learners with a Random Forest classifier.
- Developed Streamlit dashboards for students and teachers with progress tracking and analytics, and integrated the Gemini LLM API as an interactive AI tutor.
- Applied ethical AI practices by training exclusively on privacy-safe synthetic data.

### AI-Driven Recipe Generator
Python · Deep Learning · Computer Vision

- Developed a dual-model system: one for image classification (63 classes, 8,000 images) and another for ingredient generation.
- Implemented a deep learning pipeline to detect food items from images and generate relevant recipe ingredients.
- Improved ingredient prediction accuracy by leveraging a large, well-labelled dataset for classification.

## EDUCATION

**National Engineering College** *Sep 2026 – Present*
*Master of Engineering (M.E.), Computer Science*

**Indian Institute of Technology Ropar** *Jan 2025 – Jun 2026*
*Major in Artificial Intelligence*

**Francis Xavier Engineering College** *2021 – 2025*
*Bachelor of Technology (B.Tech), Artificial Intelligence and Data Science*

## CERTIFICATIONS

- **Microsoft Azure AI Essentials Professional Certificate** — Microsoft, Aug 2026 (Azure AI Studio, Microsoft Azure)
- **Career Essentials in GitHub** — GitHub, Jul 2026
- **Cybersecurity with Artificial Intelligence** · **Advanced Chatbot Development** · **Applied Machine Learning** · **Photoshop Essentials**

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, C, SQL, Dart

**AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Scikit-learn, LLM APIs (Gemini), Conversational / Voice AI, Sarvam AI, Google TTS

**Web & Backend:** Node.js, React, Vite, Socket.IO, WebRTC (mediasoup), REST APIs, HTML/CSS, Tailwind CSS, Streamlit

**Mobile & PWA:** Flutter, Firebase, OneSignal, Service Workers, Google Play Console

**Data & Cloud:** MySQL, Supabase, Firebase (Firestore, Auth, Hosting), AWS EC2

**APIs & Platforms:** WhatsApp Business Platform (Meta for Developers), Razorpay, Zoho Payments, CRM integrations

**Tools:** Git/GitHub, FFmpeg, Razorpay, Zoho Payments, coturn, Photoshop